SAPHIR: Secure Acoustic Pairing for Heterogeneous Inter-agent Rendezvous

Beyond Speech for Agent-to-Agent Dialogue on Bare Audio Channels

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Abstract

Voice AI agents that call each other still converse the wasteful way: speech → text → LLM → speech, even when both sides are machines. The viral Gibberlink demo (Feb. 2025) showed two agents can instead switch to data-over-sound tones — but with no authentication, no negotiation, no structure, and no audit trail: anyone with a speaker can spoof it. Meanwhile the emerging interop stack (MCP, A2A, ANP, InterSAGE) uniformly assumes IP connectivity and HTTPS, leaving voice-only calls, PSTN trunks, offline rooms, and noisy acoustic channels uncovered. We argue the community needs *acoustic pairing*: any two heterogeneous agents sharing nothing but a bare audio channel should detect each other, negotiate a codec, establish forward-secret encryption, and collaborate in structured tasks — while every beep stays human-inspectable. We sketch SAPHIR, a three-phase handshake (HELLO → WELCOME → KEY1 → KEY2) followed by AEAD-encrypted CBOR tasks, and report preliminary evidence from an open prototype: sub-150 B task frames, blocked spoof/replay/downgrade attacks, and a quantified robustness cliff that motivates reliability framing as the next step. We close with three research questions for the community.

1 Introduction

In February 2025 two engineers demonstrated Gibberlink [1]: two voice agents on a phone call realize both sides are AI and switch from English to `ggwave` [2] data-over-sound tones — bleeps that carry text faster than speech and skip two speech-synthesis passes. The video passed 15M views. The technical content, however, is a prompt ("switch if the other side is AI") plus raw tones: the switch is *unauthenticated*, the format is *unnegotiated*, the content is still *unstructured English*, and the exchange is *opaque to humans*. A recorded beep replays; a crafted beep injects; a legacy peer is silently beeped at.

The same year, the agent-interop community converged on a serious stack for agent collaboration: MCP for agent-to-tool access, A2A for task delegation with Agent Cards, ANP for decentralized discovery, and InterSAGE for the identity/trust substrate [3, 4]. Its threat models are maturing just as fast: A2ASecBench [5] and ASB show protocol-logic attacks bypassing default safeguards, and Dangol [6] shows even encrypted transports leak the communication graph. Yet every one of these protocols assumes URL-addressable agents over HTTP(S). None answers: *what if the only shared channel is sound?*

That gap is not exotic. Near-term agent telephony (AI receptionists, AI assistants calling businesses), PSTN trunks that mangle ultrasound, hospitals and plants where RF is restricted, and disaster/offline settings all produce voice-only encounters between heterogeneous agents. This paper's thesis: *agent pairing over bare audio deserves first-class protocol status, with security and auditability designed in — not demoed around.*

2 Vision: pairing like Bluetooth

We envision *acoustic pairing*: two agents that share nothing but a microphone and a speaker should, within seconds:

1. **Detect** mutual AI capability without trusting self-declaration alone;
2. **Negotiate** codecs and task ontologies in tens of bytes;
3. **Encrypt** with per-pairing forward secrecy, cheap enough for tones;
4. **Collaborate** in structured intents (slots, not sentences);
5. **Account** for every exchange in a human-readable sidecar log.

Properties 1–4 deliver Gibberlink's promised efficiency; property 5 answers the trust gap Cisco reports as the top blocker of agentic-AI deployment (85% exploring, 5% in production [9]): beeps the operator cannot read are a non-starter for high-stakes tasks.

3 Why the HTTP stack does not solve it

Table 1 positions acoustic pairing against the 2025–26 stack. The pattern is uniform: content standardized, transport assumed. IETF drafts on agent-protocol frameworks [8] explicitly list discovery, negotiation, and fallback among open requirements — precisely the pieces a bare-audio channel forces us to re-derive under a ∼30 Bps budget with no PKI handshake room and codecs that destroy ultrasound.

Table 1: Acoustic pairing vs. the interop stack. All prior rows assume IP.

System	Channel	Auth	Audio fallback
Gibberlink	audio only	none (prompt)	itself, insecure
MCP	HTTPS tool	yes	none
A2A	HTTPS agent	yes	none
ANP	HTTPS + DID	yes	none
InterSAGE	trust substrate	yes	none
SAPHIR (this work)	**audio / hybrid**	**eph. keys**	**native**

4 SAPHIR sketch

SAPHIR (Secure Acoustic Pairing for Heterogeneous Inter-agent Rendezvous) is a minimal pairing protocol; our prototype implements it in ∼600 lines of Python (`github.com/vaibhav-rm/SAPHIR`). Frames are length-prefixed CBOR envelopes, `u32be len` ‖ `CBOR({v,type,seq,sender,nonce,payload})`, capped at 200 B. The handshake (Fig. 1) is: **HELLO** (ephemeral `did:key`, codec list), **WELCOME** (selected codec — a peer offering no SAPHIR codec is flagged and kept in English, never silently beeped at), **KEY1/KEY2** (32 B X25519 ephemeral keys each way → HKDF-SHA256 → 32 B ChaCha20-Poly1305 session key), then **TASK/RESULT** pairs as `ChaCha(cbor({intent,slots,ontology,ts}))` with the sender bound as AAD and replays rejected by sequence plus (nonce, ct) sets. Deliberately out of scope for v0.1: PKI-backed identity, MLS groups, ultrasound modes, and retransmission.

The threat model is summarized in Table 2: the three attacks that succeed trivially against raw-tone switching are blocked by construction, while stolen-session keys and traffic analysis are declared out of scope (cf. InterSAGE [4] and [6] for the IP-side treatment).

5 Preliminary evidence

Our prototype runs the full pairing plus a hotel-booking task (check then book) with no audio hardware (byte-identical framing; over-air as next step). Measured envelope sizes: HELLO ∼194 B, KEY ∼107–113 B, TASK

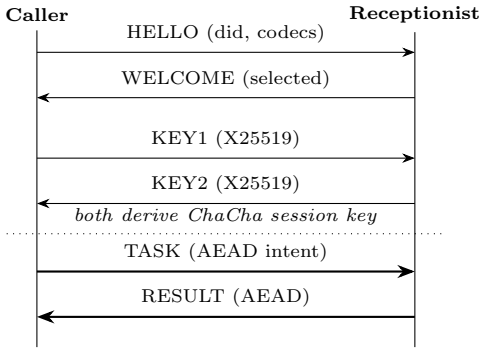


Figure 1: SAPHIR pairing: plaintext negotiation, then encrypted tasks. Every frame is sidecar-logged with its decrypted view.

Table 2: Threat model: raw tones vs. SAPHIR v0.1 (prototype-verified).

Attack	Raw tones	SAPHIR
Tone spoof / injection	succeeds	blocked (AEAD tag)
Replay of valid frame	succeeds	blocked (seq + nonce set)
Forced downgrade	silent	flagged, stays in English
Eavesdropping	plaintext	ciphertext
Stolen session keys	n/a	out of scope (needs PKI)
Who-talks-to-whom	n/a	out of scope

`hotel.check` 134 B and `hotel.book` 148 B with a compact slot encoding (`date`→`d`, `hotel.book`→`hb`; legacy full-form still decodes). A scripted attacker without session keys is rejected (`InvalidTag`); an exact replay is rejected (sequence reuse); a legacy peer is kept in English with an explicit flag — all three checks run as `evals/attacks.py`. A 50-trial/cell loss sweep (`evals/robustness.py`) shows 100% success on a clean channel, 94% at 1% frame loss degrading to 44% at 10% loss, and a sharp corruption cliff (36% at 0.1% byte-corruption, 0% at 0.5%) — expected, since AEAD must reject forgeries, and precisely the quantification that motivates ACK/retry framing next. Two genuine bugs (a shallow-copied inventory, a RESULT decoder mangling) were caught by the sweep itself.

6 Open challenges

We propose three questions for the community. **(RQ1)** What is the minimal negotiable handshake under ∼30 Bps with codecs that destroy ultrasound — can Agent-Card semantics compress into tens of bytes? **(RQ2)** How should semantic (not textual) agreement be verified over such channels, given that well-formed JSON does not imply shared meaning [7]? **(RQ3)** What does auditability mean when the channel is human-inaudible — selective disclosure of the sidecar without leaking the workflow graph [6]? Our roadmap (reliability framing, sub-140 B compaction, ephemeral-DID PKI, over-air

measurements across PCM/Opus/G.711) is documented with the artifact; we invite red-teamers.

References

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